Copernicus Marine Service

# QUALITY INFORMATION DOCUMENT

# Sea Ice Thematic Assembly Centre
# SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013

**Issue:** 2.0

**Contributors:** S. Hendricks

**Approval date by the CMEMS product quality coordination team: 15/11/2024**

# CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

| Issue | Date | § | Description of Change | Author | Validated |
|---|---|---|---|---|---|
| 1.0 | September 2023 | All | Creation of the document.<br>The previous document included many Sea Ice TAC products and was split (CMEMS-SI-QUID-011-001to007-009to015-019-V2.17) | S. Hendricks | C. Wettre |
| 2.0 | November 2024 | All | Update to new product version introduced to the service in November 2024 | S. Hendricks | C. Wettre |

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# I EXECUTIVE SUMMARY

## I.1 Products covered by this document

This document covers the Copernicus Climate Change Service (C3S) sea ice thickness climate data record (CDR) v4.0, which is brokered to the Copernicus Marine Service.

Product Name: SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013: Monthly gridded sea ice thickness from multi-sensor satellite radar altimeter data in the northern hemisphere between October and April with a resolution of 25 km for the period 1995 to 2024.

## I.2 Summary of the results

This data product is redistributed, therefore the full product quality documentation, including the Product Quality Assurance Document (Hendricks, 2024a) and the Product Quality Assessment Report (Hendricks, 2024b), are available at the C3S Climate Data Store (CDS)[1]. The present document provides a top-level summary of these reports.

The quality of the monthly gridded sea ice thickness SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013 from 1995/10 to 2024/04 has been assessed by comparison with in-situ observations and with orbit crossovers.

**Sea Ice Thickness:** Product quality indicators (bias, uncertainty and precision) vary considerably between the radar altimeter types (pulse-limited: ERS-2[2] & Envisat[3] and synthetic aperture: CryoSat-2[4] & Sentinel-3A/B[5]). The mean bias is -0.59 m for pulse-limited altimeters and -0.13 m for synthetic aperture altimeters. The uncertainty is 0.69 m for pulse-limited altimeters and 0.54 m for synthetic aperture altimeters. The mean absolute difference of low-latency (< 1 day) orbit crossovers shows a precision of 0.53 m for ERS-2, 0.37m for Envisat, 0.33 m for CryoSat-2, 0.33m for Sentinel-3A and 0.32 m for Sentinel-3B at 25 km scale.

**Important note**: Sea ice thickness is underestimated considerably (0.5 m) with respect to validation data before 2008 during the pulse-limited period. The full spatial extent of this underestimation is unknown as validation data is geographically limited, but sea ice thickness trends computed from SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013 must be considered as too low.

[1] https://cds.climate.copernicus.eu/datasets/satellite-sea-ice-thickness (last accessed: November 2024)
[2] https://earth.esa.int/eogateway/missions/ers (last accessed November 2024)
[3] https://earth.esa.int/eogateway/missions/envisat (last accessed: November 2024)
[4] https://www.esa.int/Applications/Observing_the_Earth/FutureEO/CryoSat (last accessed: November 2024)
[5] https://sentinels.copernicus.eu/web/sentinel/copernicus/sentinel-3 (last accessed: November 2024)

## I.3 Estimated Accuracy Numbers

The metrics "mean bias" and "root-mean square deviation (RMSD)" are computed from the gridded multi-sensor data, which is divided into two periods:

1. 1995 – 2010: only pulse limited radar altimeter data available (ERS-2, Envisat)
2. 2010 – 2024: synthetic aperture radar altimeter data available (CryoSat-2, Sentinel-3A/B)

The precision of the metrics is based on Level-2 trajectory data and it is available per platform (Table 1).

*Table 1: Key product quality metrics (bias, uncertainty and precision) for the two radar altimeter types (pulse-limited and synthetic aperture) in SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013.*

| **Metric** | **Radar Altimeter Type** | **Result** |
|---|---|---|
| Mean Bias (m) | Pulse-Limited [ERS-2, Envisat] (1995-2010) | -0.59 m |
| | Synthetic Aperture [CryoSat-2, Sentinel-3A/B] (2010-2024) | -0.13 m |
| RMSD / Uncertainty (m) | Pulse-Limited [ERS-2, Envisat] (2002-2010) | 0.69 m |
| | Synthetic Aperture [CryoSat-2, Sentinel-3A/B] (2010-2024) | 0.54 m |
| **Metric** | **Platform** | **Result** |
| Precision (m) | Pulse-Limited [ERS-2] (1995-2003) | 0.53 m |
| | Pulse-Limited [Envisat] (2002-2012) | 0.37 m |
| | Synthetic Aperture [CryoSat-2] (2010-2024) | 0.33 m |
| | Synthetic Aperture [Sentinel-3A] (2016-2024) | 0.33 m |
| | Synthetic Aperture [Sentinel-3A] (2018-2024) | 0.32 m |

# II PRODUCTION SYSTEM DESCRIPTION

**Production centres name:** Alfred Wegener Institute (AWI), Germany

**Production system name:** Sea ice thickness climate data record from satellite observations v4.0 produced with Python sea ice radar altimetry toolbox[6] (pysiral) v0.11 within the Copernicus Climate Change – Ocean Products contract (C3S2 312b).

**Description**

Sea ice thickness is generated from ERS-2, Envisat, CryoSat-2 and Sentinel-3A/B satellite radar altimeter data. Results from individual orbits are aggregated between October and April in the northern hemisphere for monthly periods. The output grids contain the monthly gridded data of all radar altimeter platforms in the respective month with a spatial resolution of 25 km. Sea ice thickness estimation is based on radar freeboard retrieval with satellite radar altimeter data. Conversion to sea ice freeboard and sea ice thickness is done with auxiliary information on snow mass and sea ice density. Wet snow and open melt ponds between May and September prevent retrieval. Product files are of processing Level-3 (aggregated data on spatio-temporal grids) and may contain gaps depending on the coverage of Level-2 orbit data. The full technical documentation is available on the product page[7] of the Copernicus Climate Data Store.

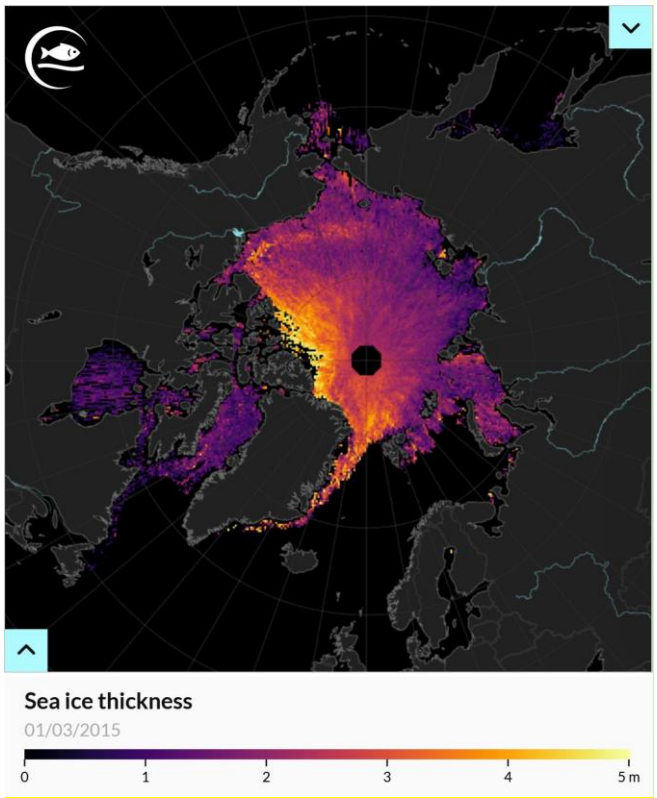


*Figure 1: Example of monthly gridded sea ice thickness (SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013-7) from CryoSat-2 data for March 2015. Visualization from Copernicus Marine Data Store*

[6] https://github.com/pysiral/pysiral (last accessed November 2024)
[7] https://cds.climate.copernicus.eu/datasets/satellite-sea-ice-thickness?tab=documentation (last accessed November 2024)

An example of the monthly gridded data is displayed in Figure 1, which contains sea ice thickness from CryoSat-2 data for March 2015 as visualized by the Copernicus Marine Data Store.

# III VALIDATION FRAMEWORK

Sea ice thickness is defined as the mean thickness of the sea ice layer in the ice-covered area fraction of each grid cell. The product quality metrics are derived from a combination of validations against independent reference and low-latency (< 24h) orbit crossovers (Figure 2). The metrics are used to estimate bias, uncertainty, precision and stability of the climate data record.

Independent reference data comprises (i) draft measurements by upward looking sonars (ULS), (ii) total thickness measurements (sea ice plus snow) by airborne electromagnetic induction (AEM) sensors and (iii) sea ice thickness from a combination of AEM and airborne snow radar data of the IceBird observation program[8]. ULS and AEM data set need to be converted to sea ice thickness. This is done with auxiliary data from the satellite sea ice thickness algorithm to ensure consistency. A full description of this methodology is given in Hendricks, 2024a. Mean bias and uncertainty from the root mean square difference (RMSD) as well as Pearson correlation coefficients are computed. The stability of the data record is estimated from the change in the validation biases over time.

In addition, low-latency orbit crossovers from Level-2 data sets are used to assess the precision of the satellites. This analysis is done per radar altimeter platform and not per radar altimeter type.

A full description of the validation methodology is given in (Hendricks, 2024a)[1]. The full documentation also includes additional information on inter-mission biases (which were low) and consistency of uncertainties from the validation exercise and algorithm uncertainty provided in the product (which were largely consistent).

[8] https://www.awi.de/en/science/climate-sciences/sea-ice-physics/projects/ice-bird.html (last accessed November 2023)

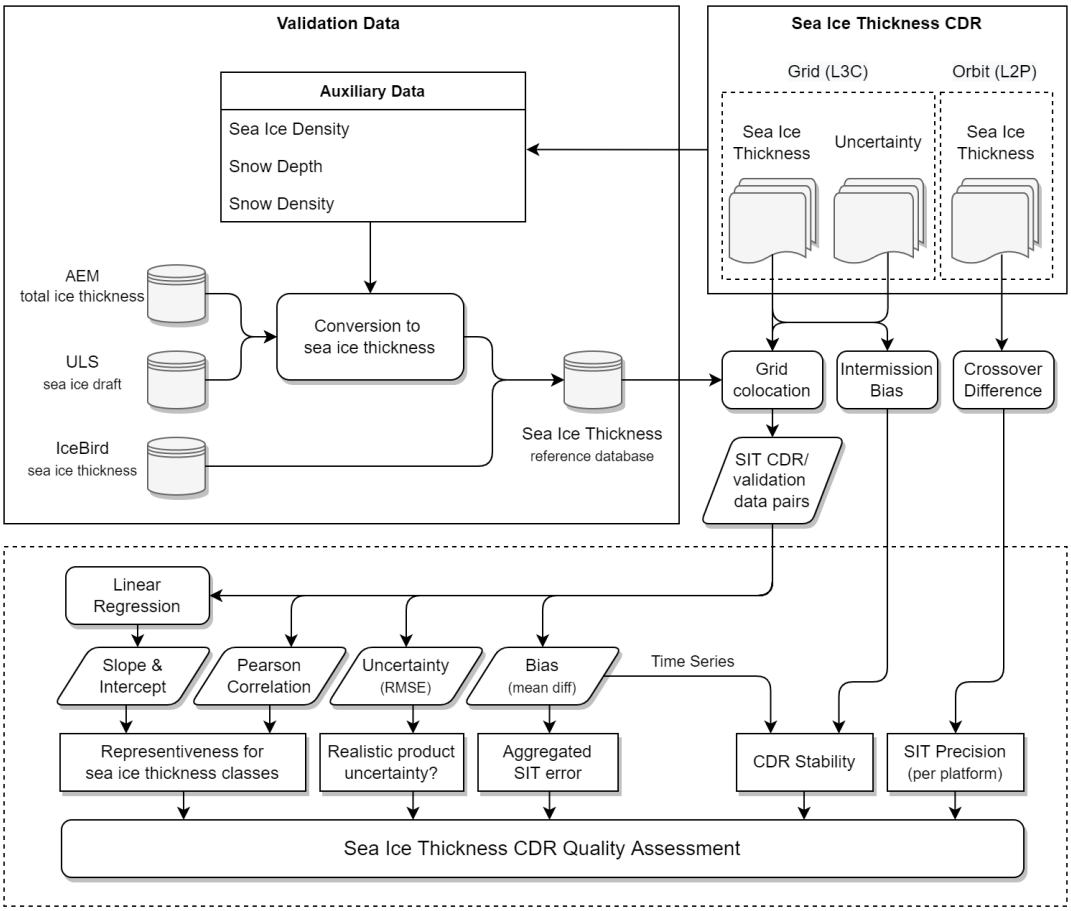


*Figure 2: Product quality assessment workflow (from Hendricks, 2024a).*

# IV VALIDATION RESULTS

A full description of the validation results is available in Hendricks, 2024b. A summary of the validation results is provided below.

## IV.1 Sea Ice Thickness

The product quality of the CDR period covered by the synthetic aperture altimeters: CryoSat-2 (2011 – 2024), Sentinel-3A (2016 – 2024) and Sentinel-3B (2018 – 2024) is significantly better in all aspects with respect to the period covered by pulse-limited altimeters: ERS-2 (1995 – 2003) and Envisat (2002 – 2011).

Table 1 lists the key product quality metrics obtained from the validation with independent reference data. Note that bias and uncertainty metrics of synthetic aperture altimeters are notably reduced compared to pulse limited altimeters, while the precision is comparable among the platforms except for ERS-2. The poorer performance of Envisat data with respect to the validation data can primarily be seen in the period before 2008 (Figure 3). During this period, validation data reveals a larger negative bias than in the earlier part of the climate data record. The reason for this bias is not fully understood, but likely related to changing sea ice surface properties due to loss of older sea ice (i.e., ice with an age of several years; Khvorostovsky et al., 2020).

*Table 1: Key product quality metrics (bias, uncertainty and precision) for the two radar altimeter types (pulse-limited and synthetic aperture) in SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013.*

| **Metric** | **Radar Altimeter Type** | **Result** |
|---|---|---|
| Mean Bias (m) | Pulse-Limited [ERS-2, Envisat] (1995-2010) | -0.59 m |
| | Synthetic Aperture [CryoSat-2, Sentinel-3A/B] (2010-2024) | -0.13 m |
| RMSD / Uncertainty (m) | Pulse-Limited [ERS-2, Envisat] (2002-2010) | 0.69 m |
| | Synthetic Aperture [CryoSat-2, Sentinel-3A/B] (2010-2024) | 0.54 m |
| **Metric** | **Platform** | **Result** |
| Precision (m) | Pulse-Limited [ERS-2] (1995-2003) | 0.53 m |
| | Pulse-Limited [Envisat] (2002-2012) | 0.37 m |
| | Synthetic Aperture [CryoSat-2] (2010-2024) | 0.33 m |
| | Synthetic Aperture [Sentinel-3A] (2016-2024) | 0.33 m |
| | Synthetic Aperture [Sentinel-3A] (2018-2024) | 0.32 m |

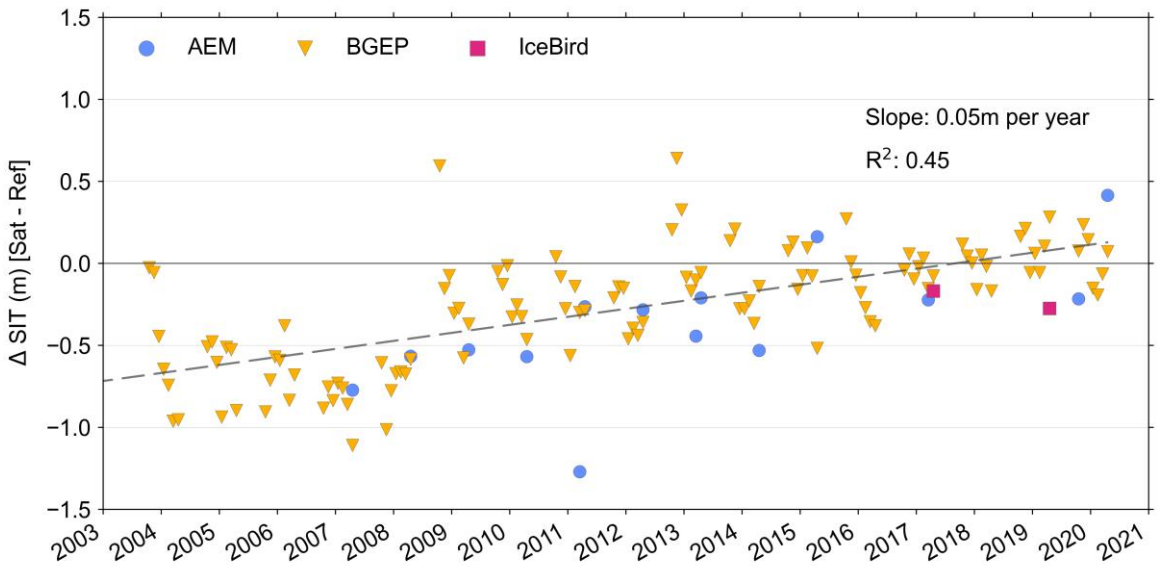


*Figure 3: Changes in monthly mean bias (for simplicity, only years for the different months are indicated in the x-axis) between satellite sea ice thickness and three reference datasets: Airborne EM (AEM), BGEP upward looking sonar data and IceBird combined airborne EM and snow radar data. The bias is computed as mean difference for monthly periods per reference dataset. The dashed line indicates the result of a trend analysis using all three reference datasets. The slope of 0.05m per year indicates that the bias drifts with time. Especially before 2009, the underestimation of sea ice thickness (negative values) by the satellite product can be observed. The later part of the data record (after 2017) is close to zero-bias (solid line). A full description of the methodology and reference datasets is given in Hendricks, 2024b.*

**Important Note**: It should be noted that the large bias for Envisat is not constant over time. The bias is considerably larger with respect to validation data before 2008 compared to the following years (Figure 3). The full spatial extent of this underestimation is unknown as validation data is geographically limited.

However, sea ice thickness trends computed from SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013 must be considered as underestimated (i.e., too low).

# V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

The product SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013 is not updated on a regular basis. The climate data record is produced once for a fixed time period. The temporal coverage of the climate data record is extended in newer versions that are released with an approximate frequency of two-three years. Each release includes quality control procedures and product validation exercise.

| Date | Change/Event description | System version | Other |
|---|---|---|---|
| 2024-10-01 | Release of new sea ice thickness CDR | V4.0 | |

# VI QUALITY CHANGES SINCE PREVIOUS VERSION

Version 4.0 of the sea ice thickness climate data record (SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013) includes data for the years 1995-2001, when radar altimeter data from ERS-2 was used, and extends the climate record until April 2024 (before it was until April 2020). Data from Sentinel-3A/B has also been added, which notably increases the density of sea ice thickness data after 2016. The monthly products in the current version use all available radar altimeter data in a given month. In contrast, in the previous version each monthly product only contained single altimeter data (v4.0: processing level L3S, v3.0: processing Level L3C).

## VII REFERENCES

Hendricks, S., (2024a) Sea Ice Thickness, Version 4.0: Product Quality Assurance Document. Copernicus Climate Change Service. Document reference C3S2_D312b_WP1-PDDP-SIT-v4.0_202406_PQAD_v4.1.

Hendricks, S., (2024b) Sea Ice Thickness, Version 4.0: Product Quality Assessment Report. Copernicus Climate Change Service. Document reference C3S2_D312b_WP2-FDDP-SIT-v4.0_202409_PQAR_v5.X.

Khvorostovsky, K.; Hendricks, S.; Rinne, E. (2020) Surface Properties Linked to Retrieval Uncertainty of Satellite Sea-Ice Thickness with Upward-Looking Sonar Measurements. Remote Sens., 12, 3094. https://doi.org/10.3390/rs12183094